

Programming Assignment 3: OOP

DISCUSSION

Comments:

My program is functioning as it is designed to do in accordance with the problem description. The driver program tests three different months with different profits (positive, negative, zero), and displays the results.

I implemented the program to use a while loop to let the user repeatedly run the program with different profit/tip inputs in the form of a menu alongside the option to view the employee statistics.

I have also implemented a feature to edit the information of employees, such as name, ID number, cuisine expertise, etc. This feature cannot change the object classification of the employees, so there will always be one owner, two chefs, and three waiters.

The database was implemented using an array of pointers to employee objects, the array is set to a fixed 6 employees, the aforementioned one owner, two chefs, and three waiters required by the assignment.

Corner Cases:

I have tested my driver program with three different cases for profit and tips, but I have also encountered an issue in that when the expected user input is an integer (i.e. profit, tip) or has a space (i.e. name), the program infinitely loops and prints to the terminal each loop.

This isn't too much of a problem for inputting new names, as that is meant to be a spaceless string, but is an error with specific inputs where the program will not work.

I have not tested:

- Extremely large positive or negative values for profit and/or tips
- Inputting invalid data types aside from strings/chars for ints

I have included the screenshots as well as the transcript of the runtime output of the program.

[Screenshots of Runtime Output of Code]

```
C:\Users\rnsot\Desktop\COMP260\Assignment 3> cmd /C "C:\Users\rnsot\.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugadapters\bin\windowsDebugLauncher.exe --stdin=Microsoft-MIEngine-In-1rwiyrjy.sy
h --stdout=Microsoft-MIEngine-Out-bhlyqan.4nd --stderr=Microsoft-MIEngine-Error-qxcpif44.3np --pid=Microsoft-MIEngine-Pid-doisteq.16z --dbgExe=C:\msys64\usr\bin\gdb.exe --interpreter=mi
EMPLOYEE DATABASE
[OPTIONS]
w - input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
s
[EMPLOYEE STATISTICS]
# 1 -----
Employee: Owner1, ID: 12, Employee Class: O
Salary: $15000
# 2 -----
Employee: Chef1, ID: 29, Employee Class: C
Salary: $10000
Expertise: French
# 3 -----
Employee: Chef2, ID: 52, Employee Class: C
Salary: $10000
Expertise: Spanish
# 4 -----
Employee: Waiter1, ID: 100, Employee Class: W
Salary: $3000
Years of Service: 2
# 5 -----
Employee: Waiter2, ID: 32, Employee Class: W
Salary: $3000
Years of Service: 3
# 6 -----
Employee: Waiter3, ID: 75, Employee Class: W
Salary: $3000
Years of Service: 0
EMPLOYEE DATABASE
[OPTIONS]
w - input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
w
This month's profit?
5000
Tips earned by Waiter1 this month?
0
Tips earned by Waiter2 this month?
100
Tips earned by Waiter3 this month?
25
EMPLOYEE DATABASE
[OPTIONS]
w - input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
s
```

```
EMPLOYEE DATABASE
[OPTIONS]
w - input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
s
[EMPLOYEE STATISTICS]
# 1 -----
Employee: Owner1, ID: 12, Employee Class: O
Salary: $18000
# 2 -----
Employee: Chef1, ID: 29, Employee Class: C
Salary: $11000
Expertise: French
# 3 -----
Employee: Chef2, ID: 52, Employee Class: C
Salary: $11000
Expertise: Spanish
# 4 -----
Employee: Waiter1, ID: 100, Employee Class: W
Salary: $3000
Years of Service: 2
# 5 -----
Employee: Waiter2, ID: 32, Employee Class: W
Salary: $1100
Years of Service: 3
# 6 -----
Employee: Waiter3, ID: 75, Employee Class: W
Salary: $3025
Years of Service: 0
EMPLOYEE DATABASE
[OPTIONS]
w - input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
w
This month's profit?
-1000
Tips earned by Waiter1 this month?
-10
Tips earned by Waiter2 this month?
20
Tips earned by Waiter3 this month?
30
EMPLOYEE DATABASE
[OPTIONS]
w - input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
s
[EMPLOYEE STATISTICS]
```

```
File Edit Selection View Go Run Terminal Help
Assignment3

EXPLORER
  OPEN EDITORS
    Employee.h
    Owner.h
    Chef.h
    Water.h
    driver.cpp
  ASSIGNMENT3
    xcode
    Chef.h
    driver.cpp
    driver.exe
    Employee.h
    Owner.h
    Water.h

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
EMPLOYEE DATABASE
[OPTIONS]
w - Input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
s
[EMPLOYEE STATISTICS]
# 1 -----
Employee: Owner1, ID: 12, Employee Class: O
Salary: $14000
# 2 -----
Employee: Chef1, ID: 29, Employee Class: C
Salary: $9000
Expertise: French
# 3 -----
Employee: Chef2, ID: 52, Employee Class: C
Salary: $9000
Expertise: Spanish
# 4 -----
Employee: Walter1, ID: 100, Employee Class: W
Salary: $3000
Years of Service: 2
# 5 -----
Employee: Walter2, ID: 32, Employee Class: W
Salary: $3000
Years of Service: 3
# 6 -----
Employee: Walter3, ID: 75, Employee Class: W
Salary: $3000
Years of Service: 0
EMPLOYEE DATABASE
[OPTIONS]
w - Input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
e
Edit which employee #?
4
New Name - Edited
New ID - 33
New Years of Service - 6
EMPLOYEE DATABASE
[OPTIONS]
w - Input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
w
This month's profit?
0
Tips earned by Edited this month?
0
Ln 111, Col 2  Spaces: 4  UTF-8  CRLF  C++  Win32
```

```
File Edit Selection View Go Run Terminal Help
Assignment3

EXPLORER
  OPEN EDITORS
    Employee.h
    Owner.h
    Chef.h
    Water.h
    driver.cpp
  ASSIGNMENT3
    xcode
    Chef.h
    driver.cpp
    driver.exe
    Employee.h
    Owner.h
    Water.h

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
w - Input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
w
This month's profit?
0
Tips earned by Edited this month?
0
Tips earned by Walter2 this month?
0
Tips earned by Walter3 this month?
0
EMPLOYEE DATABASE
[OPTIONS]
w - Input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
q
[EMPLOYEE STATISTICS]
# 1 -----
Employee: Owner1, ID: 12, Employee Class: O
Salary: $15000
# 2 -----
Employee: Chef1, ID: 29, Employee Class: C
Salary: $10000
Expertise: French
# 3 -----
Employee: Chef2, ID: 52, Employee Class: C
Salary: $10000
Expertise: Spanish
# 4 -----
Employee: Edited, ID: 33, Employee Class: W
Salary: $3000
Years of Service: 6
# 5 -----
Employee: Walter2, ID: 32, Employee Class: W
Salary: $3000
Years of Service: 3
# 6 -----
Employee: Walter3, ID: 75, Employee Class: W
Salary: $3000
Years of Service: 0
EMPLOYEE DATABASE
[OPTIONS]
w - Input profit and tips earned in $ for the month
s - show employee statistics for this month
e - edit employee information
q - quit program
q
[PROGRAM TERMINATING]
C:\Users\vnnsot\Desktop\COMPE260\Assignment 3\
Ln 111, Col 2  Spaces: 4  UTF-8  CRLF  C++  Win32
```

[Transcript of Runtime Output of Code]

```
C:\Users\rnsot\Desktop\COMPE260\Assignment 3> cmd /C
"c:\Users\rnsot\.vscode\extensions\ms-vscode.cpptools-1.19.9-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe --stdin=Microsoft-MIEngine-In-Iraviyjj.syh
--stdout=Microsoft-MIEngine-Out-bhiy3qwn.4hd
--stderr=Microsoft-MIEngine-Error-qvx3pf44.3wp --pid=Microsoft-MIEngine-Pid-doistgmq.h4z
--dbgExe=C:\msys64\ucrt64\bin\gdb.exe --interpreter=mi "
```

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

s

[EMPLOYEE STATISTICS]

1 -----

Employee: Owner1, ID: 12, Employee Class: O

Salary: \$15000

2 -----

Employee: Chef1, ID: 29, Employee Class: C

Salary: \$10000

Expertise: French

3 -----

Employee: Chef2, ID: 52, Employee Class: C

Salary: \$10000

Expertise: Spanish

4 -----

Employee: Waiter1, ID: 100, Employee Class: W

Salary: \$3000

Years of Service: 2

5 -----

Employee: Waiter2, ID: 32, Employee Class: W

Salary: \$3000

Years of Service: 3

6 -----

Employee: Waiter3, ID: 75, Employee Class: W

Salary: \$3000

Years of Service: 0

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

w

This month's profit?

5000

Tips earned by Waiter1 this month?

0

Tips earned by Waiter2 this month?

100

Tips earned by Waiter3 this month?

25

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

s

[EMPLOYEE STATISTICS]

1 -----

Employee: Owner1, ID: 12, Employee Class: O

Salary: \$18000

2 -----

Employee: Chef1, ID: 29, Employee Class: C

Salary: \$11000

Expertise: French

3 -----

Employee: Chef2, ID: 52, Employee Class: C

Salary: \$11000

Expertise: Spanish

4 -----

Employee: Waiter1, ID: 100, Employee Class: W

Salary: \$3000

Years of Service: 2

5 -----

Employee: Waiter2, ID: 32, Employee Class: W

Salary: \$3100

Years of Service: 3

6 -----

Employee: Waiter3, ID: 75, Employee Class: W

Salary: \$3025

Years of Service: 0

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

w

This month's profit?

-1000

Tips earned by Waiter1 this month?

-10

Tips earned by Waiter2 this month?

20

Tips earned by Waiter3 this month?

30

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

s

[EMPLOYEE STATISTICS]

1 -----

Employee: Owner1, ID: 12, Employee Class: O

Salary: \$14400

2 -----

Employee: Chef1, ID: 29, Employee Class: C

Salary: \$9800

Expertise: French

3 -----

Employee: Chef2, ID: 52, Employee Class: C

Salary: \$9800

Expertise: Spanish

4 -----

Employee: Waiter1, ID: 100, Employee Class: W

Salary: \$3000

Years of Service: 2

5 -----

Employee: Waiter2, ID: 32, Employee Class: W

Salary: \$3020

Years of Service: 3

6 -----

Employee: Waiter3, ID: 75, Employee Class: W

Salary: \$3030

Years of Service: 0

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

e

Edit which employee #?

4

New Name - Edited

New ID - 33

New Years of Service - 6

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

w

This month's profit?

0

Tips earned by Edited this month?

0

Tips earned by Waiter2 this month?

0

Tips earned by Waiter3 this month?

0

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

s

[EMPLOYEE STATISTICS]

1 -----

Employee: Owner1, ID: 12, Employee Class: O

Salary: \$15000

2 -----

Employee: Chef1, ID: 29, Employee Class: C

Salary: \$10000

Expertise: French

3 -----

Employee: Chef2, ID: 52, Employee Class: C

Salary: \$10000

Expertise: Spanish

4 -----

Employee: Edited, ID: 33, Employee Class: W

Salary: \$3000

Years of Service: 6

5 -----

Employee: Waiter2, ID: 32, Employee Class: W

Salary: \$3000

Years of Service: 3

6 -----

Employee: Waiter3, ID: 75, Employee Class: W

Salary: \$3000

Years of Service: 0

EMPLOYEE DATABASE

[OPTIONS]

w - input profit and tips earned in \$ for the month

s - show employee statistics for this month

e - edit employee information

q - quit program

q

[PROGRAM TERMINATING]

C:\Users\rnsot\Desktop\COMPE260\Assignment 3>
